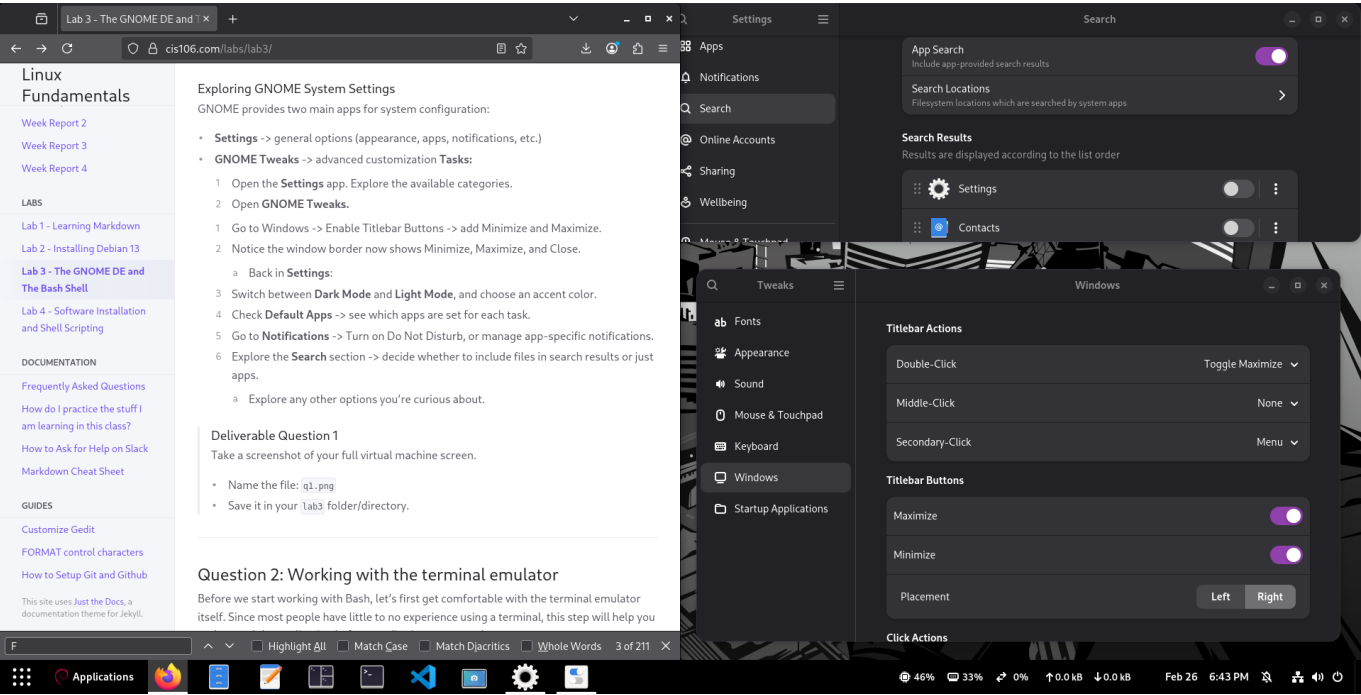
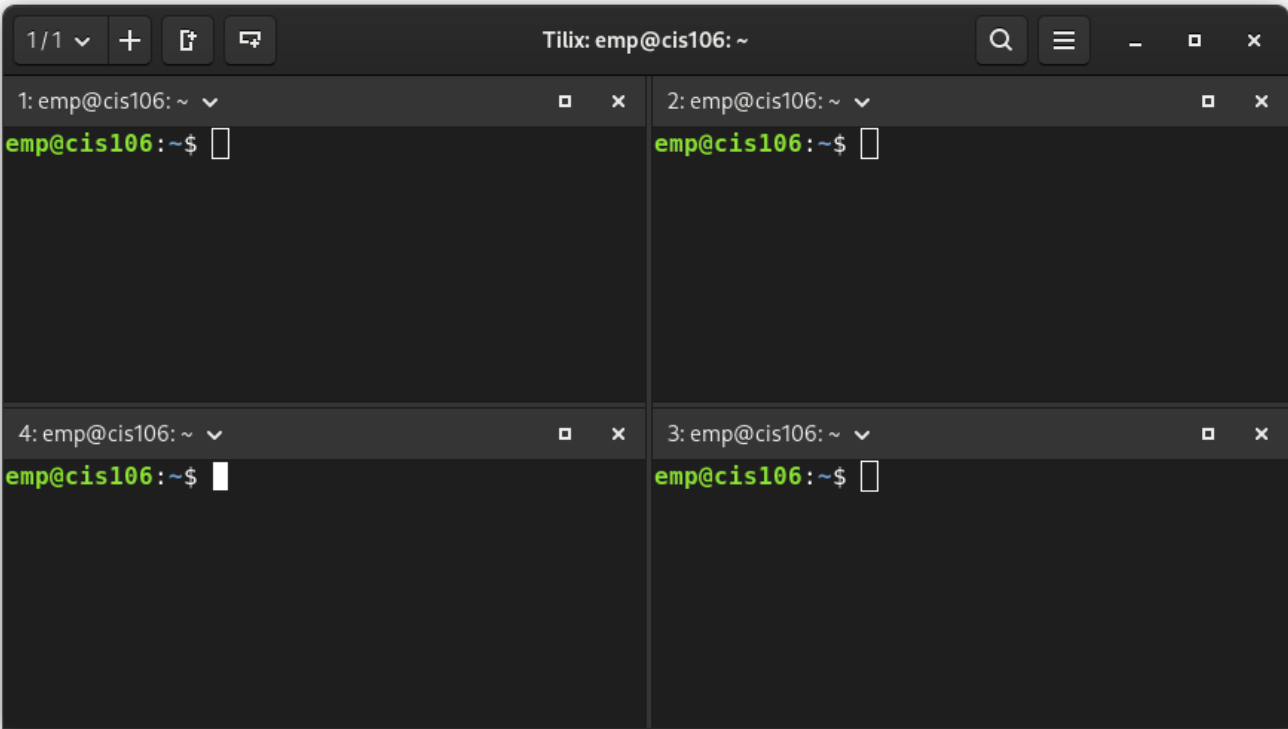


# Lab 3 Submission

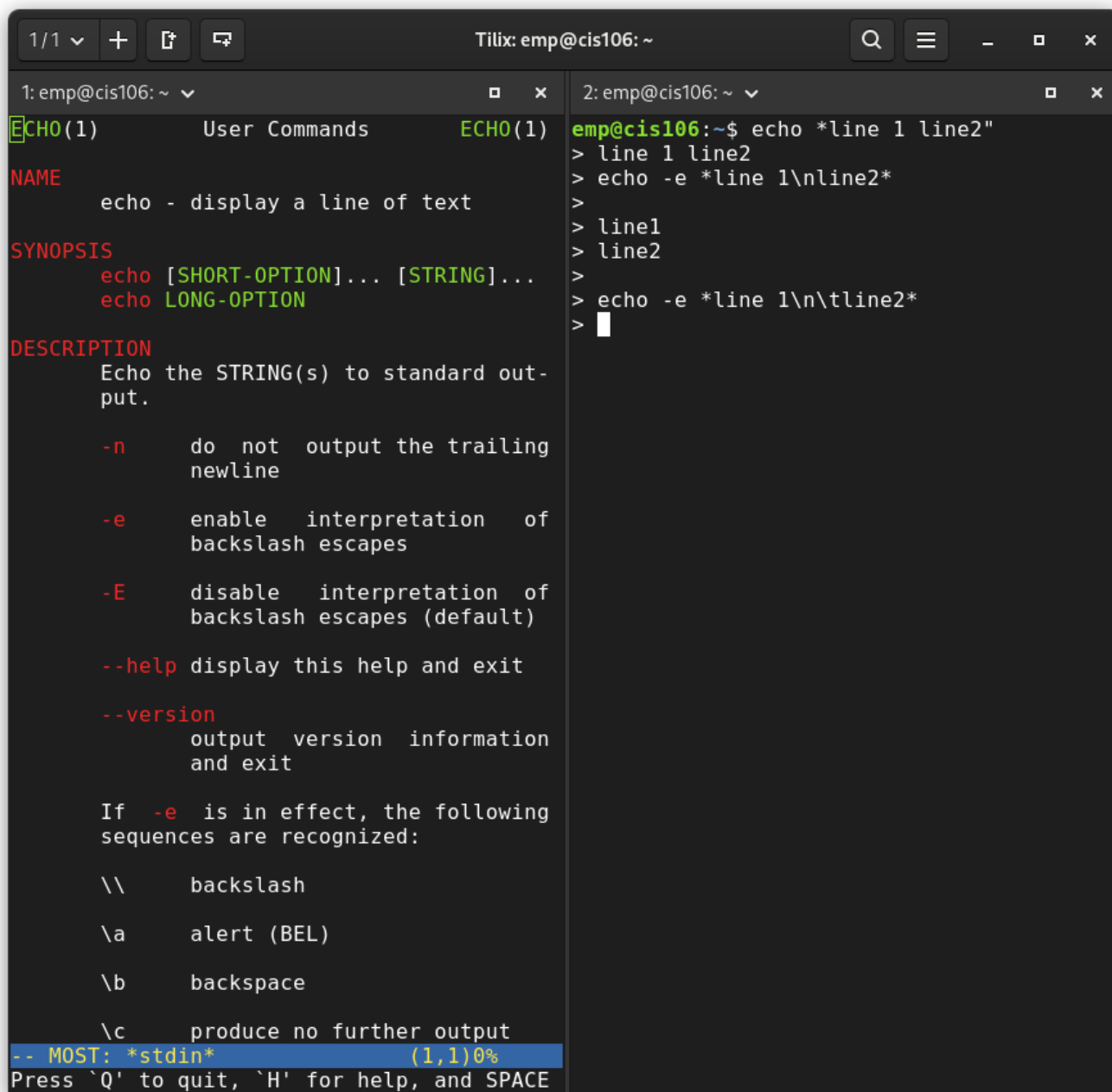
## Question 1



## Question 2



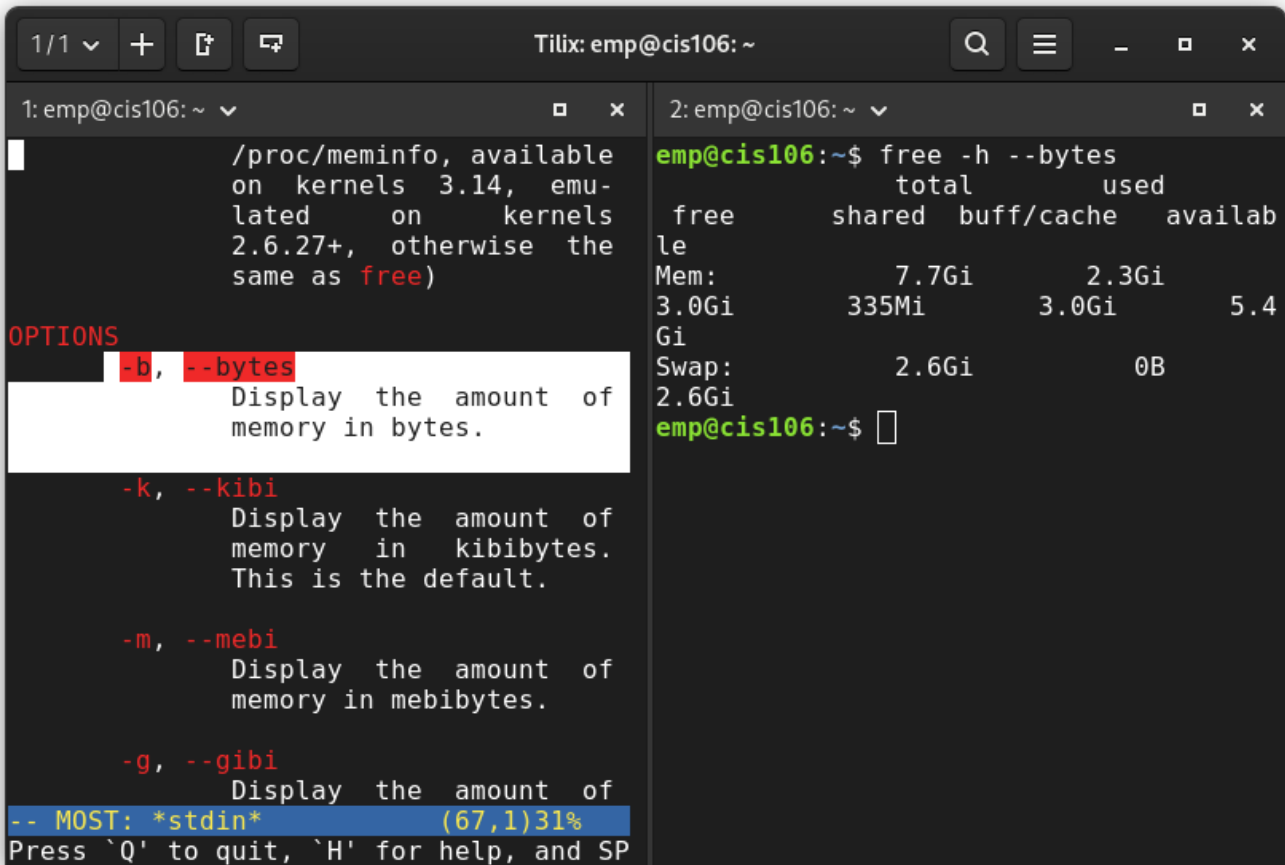
## Question 3



```
1: emp@cis106: ~  
ECHO(1) User Commands ECHO(1)  
NAME  
    echo - display a line of text  
SYNOPSIS  
    echo [SHORT-OPTION]... [STRING]...  
    echo LONG-OPTION  
DESCRIPTION  
    Echo the STRING(s) to standard out-  
    put.  
    -n      do not output the trailing  
            newline  
    -e      enable interpretation of  
            backslash escapes  
    -E      disable interpretation of  
            backslash escapes (default)  
    --help  display this help and exit  
    --version  
            output version information  
            and exit  
    If -e is in effect, the following  
    sequences are recognized:  
    \\      backslash  
    \a      alert (BEL)  
    \b      backspace  
    \c      produce no further output  
-- MOST: *stdin* (1,1)0%  
Press `Q` to quit, `H` for help, and SPACE
```

```
2: emp@cis106: ~  
emp@cis106:~$ echo *line 1 line2"  
> line 1 line2  
> echo -e *line 1\nline2*  
>  
> line1  
> line2  
>  
> echo -e *line 1\n\tline2*  
>
```

## Challenge Question



```

1/1 v + [ ] [ ] Tilix: emp@cis106: ~
1: emp@cis106: ~ v [ ] x
2: emp@cis106: ~ v [ ] x

/proc/meminfo, available
on kernels 3.14, emu-
lated on kernels
2.6.27+, otherwise the
same as free)

OPTIONS
-b, --bytes
    Display the amount of
    memory in bytes.

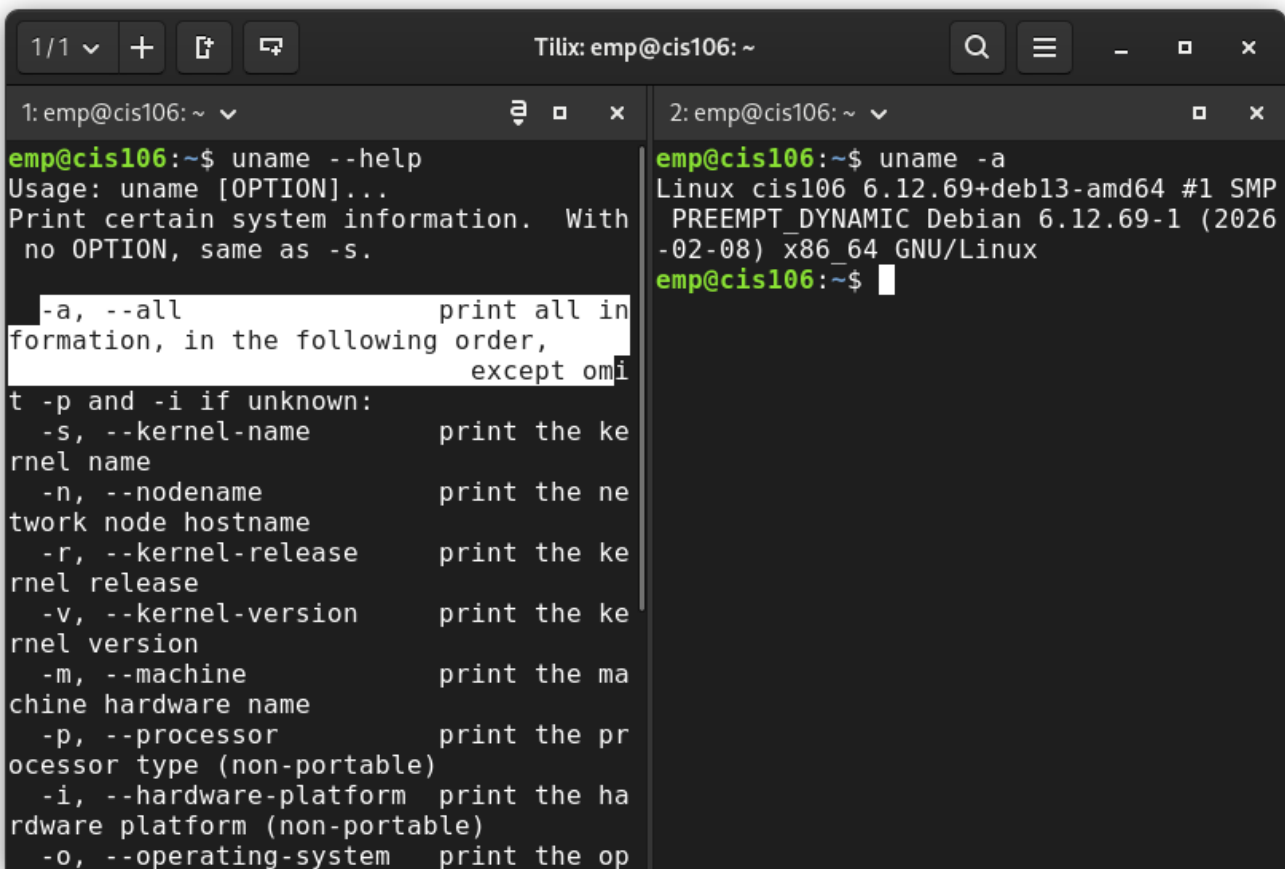
-k, --kibi
    Display the amount of
    memory in kibibytes.
    This is the default.

-m, --mebi
    Display the amount of
    memory in mebibytes.

-g, --gibi
    Display the amount of
-- MOST: *stdin* (67,1)31%
Press `Q' to quit, `H' for help, and SP

emp@cis106:~$ free -h --bytes
              total        used
free         shared buff/cache   availab
le
Mem:          7.7Gi        2.3Gi
3.0Gi        335Mi        3.0Gi        5.4
Gi
Swap:          2.6Gi          0B
emp@cis106:~$

```



```

1/1 v + [ ] [ ] Tilix: emp@cis106: ~
1: emp@cis106: ~ v [ ] x
2: emp@cis106: ~ v [ ] x

emp@cis106:~$ uname --help
Usage: uname [OPTION]...
Print certain system information.  With
no OPTION, same as -s.

-a, --all
    print all in
    formation, in the following order,
    except omi
t -p and -i if unknown:
-s, --kernel-name
    print the ke
    rnel name
-n, --nodename
    print the ne
    twork node hostname
-r, --kernel-release
    print the ke
    rnel release
-v, --kernel-version
    print the ke
    rnel version
-m, --machine
    print the ma
    chine hardware name
-p, --processor
    print the pr
    ocessor type (non-portable)
-i, --hardware-platform
    print the ha
    rdware platform (non-portable)
-o, --operating-system
    print the op

emp@cis106:~$ uname -a
Linux cis106 6.12.69+deb13-amd64 #1 SMP
PREEMPT_DYNAMIC Debian 6.12.69-1 (2026
-02-08) x86_64 GNU/Linux
emp@cis106:~$

```

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